



EPICODE

Introduction to Computer Vision

Project Overview



You are required to build a complete Computer Vision application that addresses a real-world problem. This project must demonstrate your ability to design and implement a full pipeline, including image acquisition/preprocessing, feature extraction, model implementation (classical and deep learning), and performance evaluation.

The project should be fully functional, well-documented, and hosted on a public GitHub repository.

Grading

- Practical Project: 50% of the final grade.
- Oral Examination: 50% of the final grade.



General Requirements

Project Theme

You are free to choose the theme of your application. Examples include:

- An automated traffic monitoring system (detection and tracking).
- A medical image analysis tool (segmentation and classification).
- An industrial quality control system (defect detection).
- An augmented reality application (feature matching and pose estimation).
- A generative model project (image-to-image translation).



General Requirements

Pipeline Requirements

The application must include a coherent pipeline with the following stages:

- **Data Acquisition & Preprocessing:** Handling input data, noise reduction, normalization, or augmentation.
- **Feature Engineering/Representation:** Use of handcrafted features (HOG, SIFT, etc.) or learned features (CNN/Transformer backbones).
- **Core Logic:** A classification, detection, segmentation, or generation model.
- **Post-processing:** Refining results (e.g., NMS for detection, morphological ops for segmentation).



General Requirements

Implementation Standards

- The project must be implemented in **Python** using industry-standard libraries (OpenCV, PyTorch/TensorFlow, scikit-learn).
- Code must be modular, clean, and well-commented.
- Use of pre-trained models is allowed but must be justified and combined with custom logic (e.g., fine-tuning or specialized head).



General Requirements

Performance Evaluation

You must include a comprehensive evaluation of your system using appropriate metrics:

- **Classification:** Accuracy, Precision, Recall, F1-score, Confusion Matrix.
- **Detection:** mAP (mean Average Precision), IoU.
- **Segmentation:** mIoU, Dice Coefficient.
- **Generation:** FID, Inception Score, or qualitative human-study-like analysis.



Deliverables

GitHub Repository

The repository must include:

- All source code.
- A `requirements.txt` or `environment.yml` file.
- The Technical Analysis Document in pdf (see next slide).
- A comprehensive `README.md` including:
 - Project title and overview.
 - Setup and running instructions.
 - Description of the pipeline and architecture.
 - Summary of results.
 - If Colab is used to run the code, include a link to the workbook



Deliverables

Technical Analysis Document

A PDF document (max 10 pages) containing:

- **Problem Statement:** Why is this problem relevant?
- **Methodology:** Detailed explanation of the chosen algorithms and architectures.
- **Experimental Results:** Tables and plots showing the performance.
- **Failure Analysis:** Discussion of cases where the model fails and why.
- **Ethical Considerations:** Analysis of potential biases or privacy concerns related to the project.



Final Notes

- All general requirements are mandatory. Failure to meet them will result in penalties.
- Security and ethical best practices illustrated during the course must be followed.
- Optional: Implementation of advanced features (e.g., real-time optimization, deployment as a web app/API, or using multi-modal approaches) can grant extra points.

Good luck and happy coding! 🚀



EPICODE
Institute of Technology